NLP COURSE PROJECT PROPOSAL · GRADUATE LEVEL

MemFaith

Measuring Causal Faithfulness of Memory Agents in Long-Context LLMs

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Knowledge Editing

Long-Context Agents

Mechanistic Interpretability

01

Motivation & Research Question

Memory agents such as **MemAgent** (Yu et al., 2025) process arbitrarily long documents by reading in segments and maintaining a compressed memory buffer via an *overwrite strategy*, achieving <5% performance loss on 3.5M-token QA. However, a fundamental question remains unanswered: **does the memory the agent writes actually causally drive its downstream answers?**

Dev's explainable-NLP seminar work introduced the **Editability-Faithfulness (EF) metric**, which uses knowledge editing (ROME) to test whether perturbing a rationale-cited fact flips the model's answer more than a matched control edit. On FEVER + GPT2-XL, EF $\approx$ 12.5% overall and up to 22.9% on *refutes* claims, indicating partial but noisy causal grounding.

This low EF score indicates that the vast majority of retained context does not causally drive the model's outputs, presenting an opportunity for aggressive, causal-driven context pruning.

01 (CONT.)

The Research Question

> "Can a learned discriminator accurately identify and prune irrelevant context chunks—based on their causal impact—to reduce context length while preserving downstream performance?"

This approach transforms Editability Faithfulness (EF) from a static evaluation metric into a direct training objective for behavioral distillation and efficient context selection.

02

Experimental Setup & Definitions

Memory Pressure

Defined as Total Document Tokens / Max Buffer Size

Measures how aggressively the memory system must compress or overwrite context.

Overwrite Depth K

Primary experimental variable. $K \in \{0, 2, 4, 8\}$

$K = 0$ is the direct-context baseline. Larger K means more chunking and more overwrite pressure.

Full-Context Baseline

Run the model on the complete long context to obtain the reference answer A_{full}.

All ablated runs are compared against this baseline.

Chunk Ablation

For each chunk i, remove it and rerun inference: $context - chunk_i \rightarrow A_{-i}$

If $A_{-i} \neq A_{full}$, chunk i had causal influence on the answer.

03

Data & Models

Primary Dataset: FEVER

Task: fact verification

Labels: SUPPORTS / REFUTES / NEI

Context: evidence passages + appended Wikipedia distractors

Role: controlled single-hop benchmark for chunk-level causal analysis

Secondary Dataset: HotpotQA

Task: multi-hop question answering

Context: longer evidence chains across multiple paragraphs

Role: tests whether chunk-level causal contribution is harder in multi-hop reasoning settings

Base Models

Primary: GPT2-XL (1.5B) — consistent with prior EF baseline, enables direct comparison

Extension: Qwen2.5-1.5B — optional stronger compact baseline

03 (CONT.)

Evaluation Plan

Primary Metric: Causal Chunk Score (CCS)

Secondary Metrics

- CCS vs. K degradation curve
- Stratified CCS by label: SUPPORTS / REFUTES / NEI (FEVER) and reasoning type / hop difficulty (HotpotQA)
- Full-context task accuracy (sanity check before ablation)
- Efficiency: tokens removed, runtime cost, memory reduction

Hypothesis

As overwrite depth K increases, CCS will decrease — causal contribution of individual chunks becomes diluted under stronger memory pressur

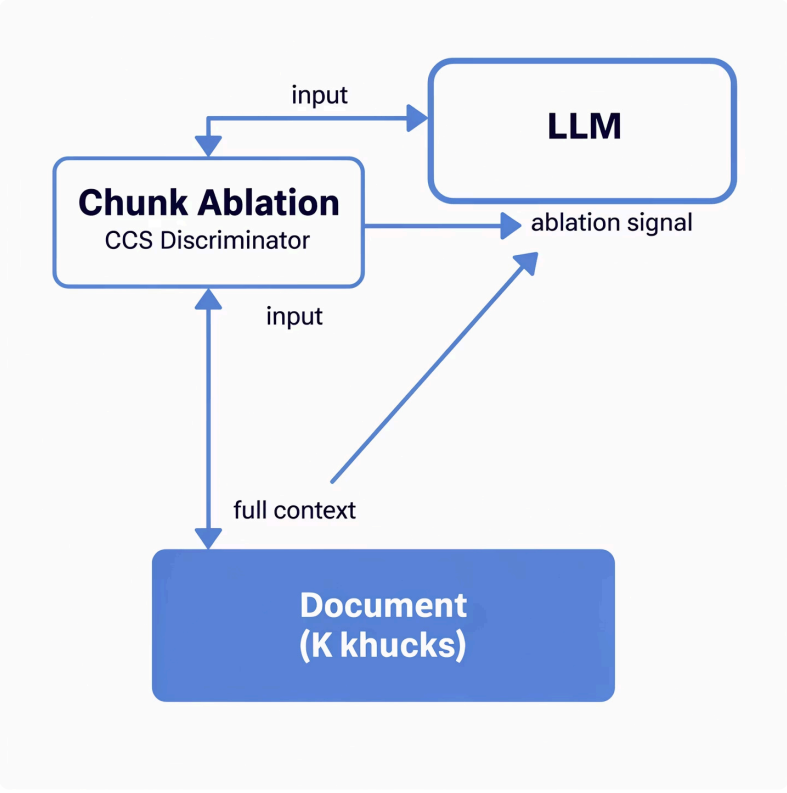
$$CCS(K) = (1/K) * \sum_i Pr[A_{-i} \neq A_{full}]$$

Higher CCS = individual chunks have stronger causal influence on the final answer.

Goal

Identify whether long-context models preserve genuinely causal evidence under increasing memory pressure.

Detailed Methodology



Causal Chunk Score

$$CCS(K) = (1/K) * \sum_i Pr[A_{-i} \neq A_{full}]$$

★ Architecture Insight

The document and the input both feed into the Chunk Ablation component — analogous to the EF discriminator in the reference diagram. The ablation signal flows up to the LLM comparison, revealing which memory chunks causally drive answers. No weight modification needed.

Aggregated flip rate across all chunk ablations at overwrite depth K

04

Evaluation Criteria

Metric	Description	Purpose
CCS Score	Used to identify relevant texts in the model context	Primary measure for discriminator training
BLEU / ROUGE	Used to compare answers syntactically	Overall architecture evaluation
LLM Evaluation	Use a pre-trained LLM to evaluate **semantic** similarity and answer correctness	Overall architecture evaluation
Degradation Curve	Plot CCS vs. K. Hypothesis: CCS degrades as K increases, as more overwrite steps dilute the causal contribution of any individual chunk.	To identify the impact of K on the model performance.

TIMELINE

Project Timeline

Week 1–2

Setup & Baselines — Implement simplified MemAgent, run GPT2-XL on FEVER full context, establish A_full baseline across K ∈ {0, 2, 4, 8}

Week 3–4

Chunk Ablation Runs — Execute leave-one-chunk-out inference for all K values on FEVER. Compute initial CCS scores.

Week 5–6

Analysis & HotpotQA — Plot CCS vs. K degradation curve, stratify by label. Extend ablation runs to HotpotQA for multi-hop comparison.

Week 7–8

Evaluation & Writeup — Compute all secondary metrics (accuracy, efficiency). Run Qwen2.5-1.5B extension if time permits. Begin paper draft.

Week 9

Final Presentation — Polish results, finalize figures, prepare slides and demo.

Project Summary

We train a lightweight discriminator using causal-intervention-based EF scores as supervision, enabling highly efficient inference across long contexts.

Causal-Driven Context Pruning

A novel framework that transforms Editability Faithfulness (EF) from an evaluation metric into a training objective for context selection.

Behavioral Distillation

The student discriminator learns to predict the "causal weight" of chunks, mimicking the teacher model's behavioral change under intervention.

Performance vs. Faithfulness

By filtering low-EF chunks, the system aims to reduce inference latency while ensuring the model remains grounded in key factual evidence. This is different from the traditional RAG because we're moving from semantic relevance to causal necessity.

05

Limitations

Label Generation Bottleneck

Generating gold EF labels requires expensive model editing (e.g., ROME) for every chunk, creating a massive computational overhead during the offline phase.

Independence Assumption

Current single-chunk ablation ignores "synergistic relevance," where multiple chunks are individually weak but collectively essential for an answer.

Extraction Artifacts

The quality of the discriminator is capped by the triple extractor; spurious extractions can lead to noisy or misleading EF scores.

Single domain

FEVER covers short, single-hop facts; multi-hop faithfulness under memory overwrite is untested.

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Future Work

Multi-Chunk Interaction Modeling: Develop set-aware discriminators that can capture the joint causal influence of overlapping or redundant chunks.

Lightweight Teacher Signals: Investigate if simpler methods (like gradient-based attribution or attention-masking) can proxy for full EF editing to speed up training.

Cross-Task Generalization: Evaluate the EF-trained discriminator on non-fact-checking tasks, such as multi-document summarization or complex RAG scenarios.

References & Connections

- ## Yu et al., 2025

  MemAgent - sliding-window memory agent achieving <5% performance loss on 3.5M-token QA via overwrite strategy.

- ## Explainable-NLP Work

  Editability-Faithfulness (EF) metric - FEVER + GPT2-XL baseline. EF $\approx$ 12.5% overall, up to 22.9% on *refutes* claims.

- ## ROME

  Rank-One Model Editing - parametric edit method used to modify internal factual beliefs in GPT2-XL and comparable ~1.5B models.

- ## FEVER Dataset

  Fact Extraction and VERification benchmark. Dev set with SUPPORTS / REFUTES / NEI labels, augmented with Wikipedia distractor passages.